

EX.NO : 09

IMPLEMENTING ERROR DETECTING CODE USING PARITY CHECK

AIM :

To write a Java program for Error Detecting code using Parity Check.

ALGORITHM :

- 1) Start the program.
- 2) Input the size of the message.
- 3) Input the message as bits.
- 4) Calculate the parity bit based on the count of set bits in the message.
- 5) Output the modified message bits with the appended parity bit.
- 6) Stop the program.

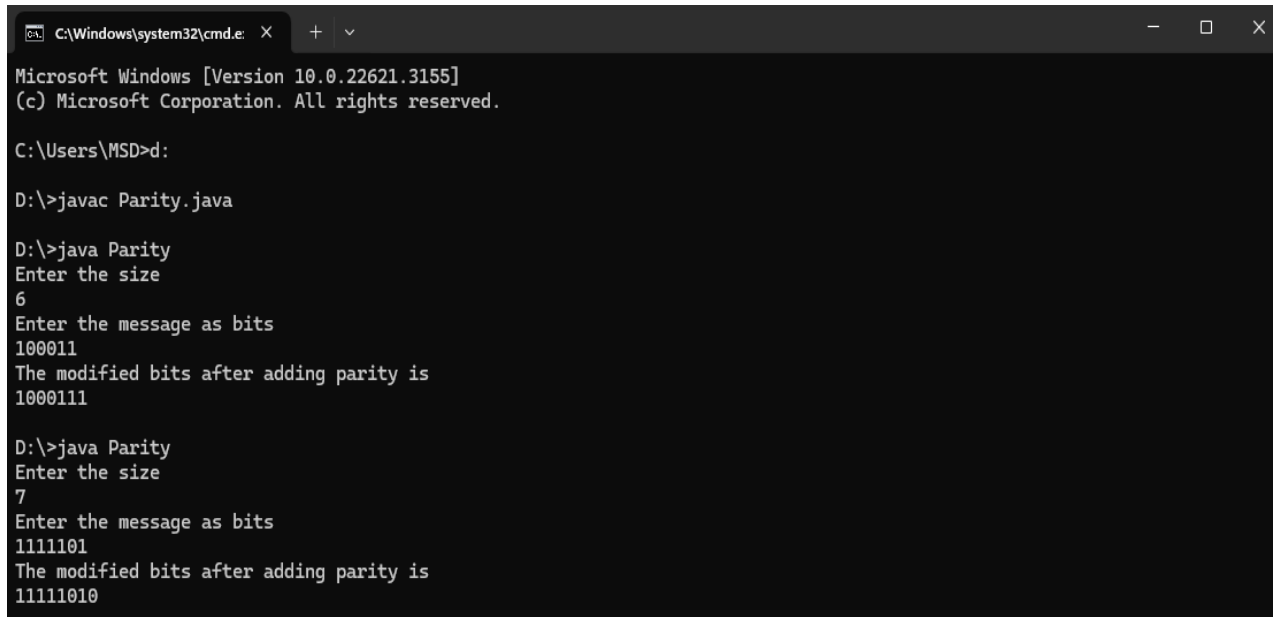
PROGRAM :

```
import java.util.*;

class Parity
{
    public static void main(String[] args)
    {
        Scanner in = new Scanner(System.in);
        System.out.println("Enter the size");
        int size = in.nextInt();
        System.out.println("Enter the message as bits");
```

```
String mess = in.next();
int[] arr = new int[size + 1];
for (int i = 0; i < size; i++)
    {
        arr[i] = mess.charAt(i) - '0';
    }
int count = 0;
for (int i = 0; i < size; i++)
    {
        if (arr[i] == 1)
        {
            count++;
        }
    }
arr[size] = (count % 2 == 0) ? 0 : 1;
System.out.println("The modified bits after adding parity is");
for (int i = 0; i < size + 1; i++)
    {
        System.out.print(arr[i]);
    }
System.out.println();
}
```

OUTPUT :



```
C:\Windows\system32\cmd.exe X + v
Microsoft Windows [Version 10.0.22621.3155]
(c) Microsoft Corporation. All rights reserved.

C:\Users\MSD>d:

D:\>javac Parity.java

D:\>java Parity
Enter the size
6
Enter the message as bits
100011
The modified bits after adding parity is
1000111

D:\>java Parity
Enter the size
7
Enter the message as bits
1111101
The modified bits after adding parity is
11111010
```

RESULT :

Thus, the program for Error Detecting code using Parity Check is successfully executed and the output is verified.